# FIBRE PRODUCTS OF DECENT, REASONABLE OR PROPERTY (BETA) SPACES INHERIT THE PROPERTY

THE STACKS PROJECT AUTHORS

*Editorial note.* Counted span is the author's own proof environment, source lines 4046-4053, immediately following the statement at 4038-4044.

*Editorial note.* Three same-chapter line-indexed stubs are recorded in assembly.external_references.

*Allowed standard prerequisites (editorial).*

- Source property transfers to the morphism. Lemma lemma-property-for-morphism-out-of-property of this chapter, cited at proof lines 4046-4047. X decent/reasonable/(beta) makes the morphism X -> Y carry the property.
- The property descends along base change of the morphism. Lemma lemma-base-change-relative-conditions of this chapter, cited at proof lines 4048-4049. Base change X x_Y Z -> Z carries the property.
- Morphism property transfers back to the source space. Lemma lemma-property-over-property of this chapter, cited at proof lines 4050-4053. A space admitting a morphism with the property towards it carries the property itself.

*Source reference.* Original lemma 03L0: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 3953–3953. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 03JX: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 952–952. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 0ABX: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 3936–3936. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 0ABY: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 3963–3963. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

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*Date*: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

**Lemma 0.1.** *Let $S$ be a scheme. Let $f : X \to Y$, $g : Z \to Y$ be morphisms of algebraic spaces over $S$. If $X$ and $Z$ are decent (resp. reasonable, resp. have property $(\beta)$ of Lemma 03JX), then so does $X \times_Y Z$.*

*Proof.* Namely, by Lemma 0ABX the morphism $X \to Y$ has the property. Then the base change $X \times_Y Z \to Z$ has the property by Lemma 03L0. And finally this implies $X \times_Y Z$ has the property by Lemma 0ABY. □